

Energy-Guided Sampling for Controllable Text Generation

From Langevin Dynamics to Amortized Inference

Sarthak Singh

Institut für Maschinelle Sprachverarbeitung, Universität Stuttgart

Examiners: Prof. Dr. Ngoc Thang Vu Dr. Antje Schweitzer

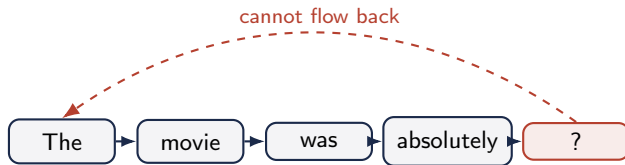
Supervisor: Prof. Dr. Ngoc Thang Vu

M.Sc. Computational Linguistics

Thesis Defense

The problem autoregressive models cannot solve natively

- Autoregressive generation: one token at a time, left to right.
- A token is fixed the moment it is emitted.
No mechanism to revise an earlier token in light of a later one.
- Global properties (sentiment, factual constraint) live in the *whole* sequence.



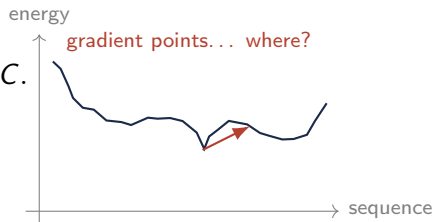
Control means steering the *completed* object. That is a global problem.

The energy-based promise, and its one assumption

Treat the frozen likelihood as an **energy** over sequences and sample from it:

$$p(\mathbf{x}) \propto \exp(-E(\mathbf{x})), \quad E = -\log p_{\text{LM}} + \lambda C.$$

- Add any constraint C at inference time.
- One frozen model, any objective.
Plug and play.



A jagged landscape: the local slope need not lead anywhere good.

The untested assumption:

the gradient of the frozen likelihood is a usable search direction on discrete text.

Research questions

- RQ1 Does the *input-embedding* gradient give a useful proposal direction for token revision, and if not, which local quantities do?
- RQ2 What does the Metropolis–Hastings correction do in each sampler?
- RQ3a Does a GFlowNet on this reward learn a policy that generates high-reward text?
- RQ3b Does its tuned energy become more navigable?
 - E (extension) Can an additive constraint steer generation here?

Three candidate causes, eliminated in order: A the target, B the model, C the derivative and the conditioning.

The proposal the main grid actually draws from

Before any ablation: how sharp is the proposal?

Measured entropy, flagship config

step 0: 10.8248 nats

floor over 50 steps: 10.28

uniform ceiling $\log |V| = 10.8249$

- Effective support $\approx 29,000$ tokens at its *sharpest*.
- Gradient term contributes ≈ 0.009 nats of logit spread.

The calibrated proposal is numerically uniform.

Proposal	exact	KL
gradient (Langevin)	0.0%	6.541
uniform draw (control)	0.5%	6.538

A uniform proposal, in a separate implementation, reproduces the flagship sampler to three significant figures.

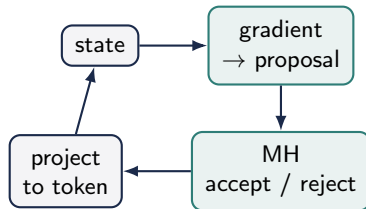
So the null was re-tested

a 5×5 sweep over step size and temperature, spanning entropy $10.82 \rightarrow 0.000$. The null holds in every cell; recovery never exceeds 2%.

Two samplers, implemented faithfully

- **DLS** (discrete): stays in token space; gradient only builds a proposal.
- **CLS** (continuous): relaxes into embedding space; projects back to tokens.
- **Metropolis–Hastings** correction, applied faithfully.
- Step size set by **oracle calibration** ($\epsilon : 10.5 \rightarrow 0.1$).

5 energies, 145 configurations, paired seeds, equivalence margin fixed in advance.



the step t
skips

One sampler loop, run to convergence, not stopped early.

The central result: gradient vs. a norm-matched random direction

A three-way ablation isolates the gradient's *direction* from its *magnitude*.

Paired contrast, $n = 1000$

last iterate	+0.133	$[-0.063, +0.326]$
chain mean	+0.136	$[-0.012, +0.287]$
chain minimum	+0.033	$[-0.079, +0.145]$

All three inside the ± 0.327 margin fixed in advance: equivalence certified, not merely non-significant.

The gradient direction is equivalent to noise of the same norm.

Energy	policy	random
GPT-2 Large	6.474	6.430
GFN lb0-500	6.009	6.001
GFN lb0-2000	6.698	6.552
GFN lb1-500	6.008	5.338

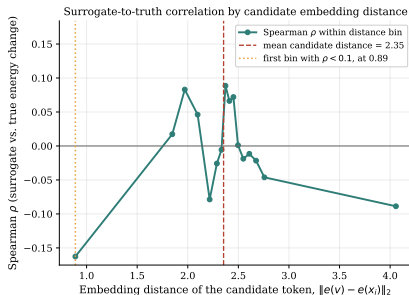
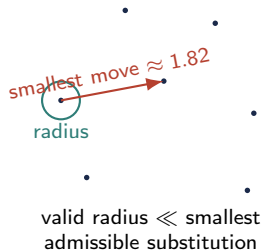
Final KL (lower better). Differences small and inconsistent in sign.

Why, part 1: the linearization fails before the first move

The proposal ranks tokens by a first-order surrogate. It is uncorrelated with the true energy change: $|\rho| < 0.06$ on all five energies ($n = 400,000$).

- Smallest token move: 1.82 (nearest-neighbour spacing).
- The surrogate is already uninformative at that distance.

The gradient is inapplicable, not merely imprecise.



Why, part 2: the correction breaks the continuous sampler

Decompose the acceptance ratio for boundary-crossing moves:

The two terms

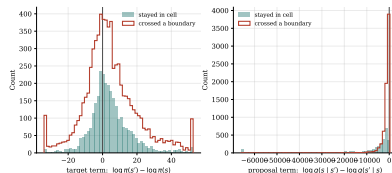
target log-ratio = +4.60 (good moves)
proposal log-ratio = -1325 (kills them)

- The projected drift is not Lipschitz across a cell boundary.
- CLS accepts within-cell moves $\approx 0\%$; DLS, 100%.

The exact correction disables CLS; it rescues DLS.

Exact for the proposal it corrects. What fails is the regularity underneath.

Metropolis-Hastings acceptance ratio, target and proposal terms, by whether the proposal crossed a Voronoi cell boundary



The target is not what fails the recovery

Score the *same* energy without its gradient (GPT-2 Large, masked recovery):

Method (no gradient of the energy)	mean KL
Ground-truth fill (floor)	0.00
Best Langevin configuration (uses the gradient)	~ 6.4
Gradient-free Gibbs sampler	6.69
Top- k rescore, one forward pass	4.43

Hypothesis A rejected: the energy is searchable, just not by this derivative.

Scoped to *this* task. As an objective to *maximize* in free generation the same energy is degenerate (backup: the likelihood trap).

GFlowNet amortization does not escape the failure

A GFlowNet only ever *evaluates* the reward, never differentiates it.

- Three failure modes: **capacity collapse**, **length collapse**, **reward hacking**.
- Tuning moved the energy by 31–57 nats. . .
- . . . yet Langevin sampling on the tuned energy is *indistinguishable* from the untuned base.

Changing the model through this objective did not make the local surrogate useful.

Policy-level (RQ3a) and energy-level (RQ3b) are separate findings, and both are negative.

The exact case: the final-token gradient is zero

Under causal attention with `inputs_embeds`, the gradient of the sequence log-likelihood w.r.t. the *final* token's input embedding is

$$\nabla_{\mathbf{e}_L} \log p_{\text{LM}}(\mathbf{x}) = 0 \quad (\text{exactly}),$$

while the energy *difference* between candidate final tokens is exactly the model's conditional log-ratio. Energy maximally usable; gradient provably useless.

Position	LM grad norm
final token	0.0000
second-to-last	> 0
middle	> 0

Langevin recovers 0% at all three positions; exact-energy methods 18–55%.

The headline: same model, same sampler, different derivative

A token enters the likelihood *twice*: as the input that predicts the suffix, and as the *target* of the previous position's prediction. The input-embedding gradient sees only the first.

$$\frac{\partial \tilde{L}}{\partial \mathbf{z}_i[v]} = \underbrace{\log p(v \mid \mathbf{x}_{<i})}_{\text{self, discarded}} + \underbrace{\mathbf{g}^\top \mathbf{e}(v)}_{\text{future}}$$

Relaxed token-indicator derivative: the self term *added to* the same future term, not in place of it. One forward pass, frozen weights.

At the calibrated $T = 5$ even this proposal falls to 2%: the temperature, not the model, was the binding constraint.

Surrogate	ρ (near)	exact
input embedding	0.03	0.0%
token indicator	0.73	40.0%

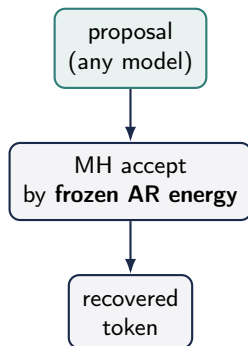
GPT-2 Large. Llama-3 8B: 41.0% against 0.0%. Nothing retrained; only the derivative changed.

0 $\rightarrow$ 40%

The conditioning ladder: what a proposal may look at

One Metropolis chain, one exact AR energy, same 200 sequences. *Only the proposal changes.*

Proposal	conditioning	exact
input-emb. gradient	none	0.0%
norm-matched random	none	0.0%
uniform draw	none	0.5%
AR conditional	left only	23.5%
SEDD (score-trained)	both sides	39.0%
RoBERTa MLM (no score obj.)	both sides	44.5%



Gradient-free comparator on the same task: top- k rescore 33%, Gibbs 18.5%. So 39% is a gain over 33, not over nothing.

Score training is sufficient, not necessary. An ordering, not an isolation (backup).

Takeaways

- ❶ The *input-embedding* gradient is equivalent to a norm-matched random direction, certified over 1000 paired sequences at a margin fixed in advance.
- ❷ Not the target on this task, and not an absence of information in the model: the same frozen weights, differentiated in token-indicator coordinates, recover 40% (GPT-2) and 41% (Llama-3).
- ❸ What buys the rest is right-context access, not a score objective: RoBERTa 44.5% > SEDD 39% > AR conditional 23.5%.

Differentiate the right object, or propose from the output side.

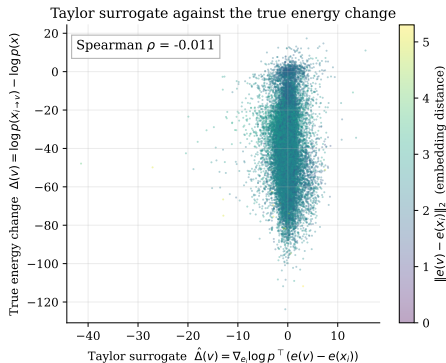
Not: evaluate rather than differentiate. The derivative was in the wrong coordinates.

Thank you.

Questions?

Backup: the linearization radius and self-term blindness

- The surrogate correlates with the *future* term ($\rho = 0.03$), never the *self* term ($\rho = -0.10$).
- Mean absolute self term 15.0 nats; future term 24.2 nats.
- The input-embedding gradient is structurally blind to the candidate's own left-context fit, which enters as a discrete index.



Backup: the Metropolis–Hastings chain

$$\alpha = \min \left\{ 1, \frac{p(\mathbf{s}') q(\mathbf{s} \mid \mathbf{s}')}{p(\mathbf{s}) q(\mathbf{s}' \mid \mathbf{s})} \right\}$$

- Target ratio: prefers good states. Proposal ratio: enforces reversibility.
- MALA needs a Lipschitz drift (Roberts and Tweedie 1996; Dalalyan 2017; Durmus and Moulines 2017).
- The projected target is piecewise constant; the differentiable pathway's drift jumps across a cell boundary, so the reverse density vanishes.

Backup: likelihood is a poor objective, yet used everywhere

- The **likelihood trap**: the lowest-energy text is the most degenerate (Holtzman 2020, reproduced here). Within-strategy repetition correlation up to +0.91.
- Resolution: likelihood ranks single-token substitutions well (used by the KL metric and Gibbs), but maximizing absolute sequence likelihood in free generation is degenerate.
- So using it as the KL metric, the reward, and the SEDD comparison is consistent: relative-to-reference fit, not absolute likelihood as quality.

Backup: MuCoLa / COLD, and why they appear to work

- The continuous sampler follows the COLD/MuCoLa mechanism faithfully, including the correction those works typically omit.
- Without the exact correction the update, *as reimplemented and measured here*, behaves as an early-stopped biased optimizer rather than a sampler from the stated energy. That is a statement about the mechanism, not a reinterpretation of those systems' published results, whose tasks, tuning and evaluation this study did not reproduce.
- The constrained *mucoLa* arm is that comparison, run at the energy level (final KL under the same base model), not the task level.

Backup: task design and corpus

- Masked-token recovery on WikiText-2 validation; the base GPT-2 Large is fine-tuned on ROCStories (out-of-domain test).
- In-domain ROCStories diagnostics reproduce the null and the likelihood trap: not a domain-shift artifact.
- The KL metric compares the recovered token's next-token conditional against the ground-truth-conditioned reference; synonyms score well.
- Llama-3 is a cross-architecture control, never on a shared numerical axis. One Llama contrast is nominally significant against the gradient, but it is one of dozens uncorrected, is measured with the correction off, and does not reproduce on GPT-2 sharp cells; the supported claim is *indifference*, not anti-guidance.

Backup: why did the temperature flatten everything?

- Proposal logit = $\left[\underbrace{\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{e}(x_i))}_{t_2} - \underbrace{\frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i)\|^2}_{t_1} \right] / T$.
- At the calibrated cell $T = 5$ divides a spread of a few nats into a 10.82-nat entropy budget: the softmax is uniform whatever the surrogate says.
- Sharpness from the *distance* term is sharp about *staying put* ($t_2/t_1 = 1.8$); sharpness from the *surrogate* is sharp about the ranking ($t_2/t_1 = 229$). Only the second tests the direction.
- The recovering cell needs large ϵ (kills t_1) and low T together.

Backup: what exactly is the token-indicator derivative?

Relax the token indicator $\mathbf{z}_i$ from its simplex vertex, in *both* roles:

$$\tilde{L}(\mathbf{z}_i) = \sum_v \mathbf{z}_i[v] \log p(v \mid \mathbf{x}_{<i}) + L_{\text{future}}(\mathbf{E}^\top \mathbf{z}_i)$$

- First term: the current position's own log-likelihood with the discrete *target* relaxed. Second term: the suffix, with the relaxed indicator mapped through $\mathbf{E}$ to an input embedding.
- At a vertex $\tilde{L}$ equals the sequence log-likelihood, so this is a relaxation of the same object.
- It is *not* $\mathbf{E}^\top \mathbf{g}$, the ordinary one-hot input gradient through the embedding lookup: that carries the future term only. The whole difference is the self term.

Backup: why does RoBERTa beat SEDD?

- Both are bidirectional; only SEDD is score-trained. RoBERTa 44.5% vs SEDD-medium 39.0%, so the score objective is not what buys recovery.
- RoBERTa-large is bigger and trained on more text than SEDD-small/medium, and masked-token prediction *is* its native query at any noise level, whereas SEDD answers it at a schedule point.
- So the claim is an *ordering by conditioning*, not an isolation of it and not a ranking of the two models: the arms differ in scale, corpus and architecture as well, and only the tokenizer and the chain are held fixed across all of them. The SEDD arms are pilots.
- Tokenizer bridge: 99.95% of RoBERTa ids map to one GPT-2 token; no ground-truth token fell outside it. Uniform control through the same code: 0.5%, so the pipeline leaks nothing.

Backup: why a GFlowNet, and the task-comparability caveat

- A GFlowNet samples in proportion to reward (not the mode), and never differentiates it, so a gradient pathology would be invisible to it.
- It reformulates infilling as left-to-right generation under a restructured prompt; the samplers do in-place bidirectional substitution.
- The comparison is therefore run at the energy level, not the task level; the left-to-right reformulation is not adopted for the samplers because it would dissolve the revision capability under study.